

$$\underline{P_0 + S_0} = \underline{C_0 + ke^{-rT}}$$

$$\underline{P_0 + S_0 - C_0} = \underline{ke^{-rT}}$$

↑
portfolio

at T → k
at 0, → ke^{-rT}

at T

$$\max(\underline{k - S_T}, 0)$$

$$+ S_T - \max(\underline{S_T - k}, 0)$$

$$S_T > k \quad \swarrow \quad \searrow \quad S_T \leq k$$

$$0 + S_T - (S_T - k) = k$$

$$k - S_T + S_T - 0 = k$$

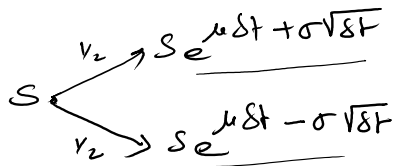
stock = Ginger
bond = Gardiz } → Mosala ↗ ↘

\$

$$\begin{aligned}
 \pi_1^- &= \phi_1 s_1 + \psi_1 \underline{\underline{B_1}} \\
 &= B_1 (\phi_1 s_1 \cdot B_1^{-1} + \psi_1) \\
 &= B_1 (\phi_1 B_1^{-1} s_1 + (E_0 - \phi_1 B_0^{-1} \cdot S_0)) \\
 &= B_1 (\underline{\underline{E_0}} + \phi_1 (s_1 B_1^{-1} - S_0 B_0^{-1})) \\
 &= B_1 (\underline{\underline{E_0}} + \phi_1 \cdot (\Delta Z_1)) = B_1 (\underline{\underline{E_1}})
 \end{aligned}$$

$$\psi_{i+1} = (E_i - \phi_{i+1} \underbrace{B_i^{-1}}_{Z_i} s_i)$$

$$\begin{aligned}
 \underline{\underline{E_0 B_0}} &= \phi_1^f S_0 + \psi_1^f B_0 \leftarrow \\
 E_0 &= \mathbb{E}^Q \left(\frac{x}{B_T} \mid \mathcal{F}_0 \right) \\
 E_i &= E_0 + \sum_{k=1}^i \phi_k \cdot (\Delta Z_k)
 \end{aligned}$$



$$0 \rightarrow t$$

$$(S_0) \quad (S_t)$$

$X_n \rightarrow \# \text{ up moves}$
 $n - X_n \rightarrow \# \text{ down moves}$

$$S_t = S_0 \times \left(e^{\mu \delta t + \sigma \sqrt{\delta t}} \right)^{X_n} \times \left(e^{\mu \delta t - \sigma \sqrt{\delta t}} \right)^{n - X_n}, \quad X_n \sim \text{Binom}(n, 1/2)$$

$$= S_0 e^{\mu(n\delta t) - \sigma n \sqrt{\delta t}} \times e^{2\sigma \sqrt{\delta t} \cdot X_n}$$

$$= S_0 e^{\mu t - \frac{\sigma n}{\sqrt{n}} \sqrt{t}} \times e^{\frac{2\sigma}{\sqrt{n}} \times \sqrt{t} \cdot X_n}$$

$$S_0 \cdot \left(\frac{e^{\mu \delta t + \sigma \sqrt{\delta t}}}{e^{\mu \delta t - \sigma \sqrt{\delta t}}} \right)^{X_n} \times \left(e^{\mu \delta t - \sigma \sqrt{\delta t}} \right)^n$$

$$e^{2\sigma \sqrt{\delta t}}$$

$$S_t = S_0 \exp \left(\mu t + \sigma \sqrt{t} \left(\frac{2(X_n) - n}{\sqrt{n}} \right) \right)$$

$$\frac{2X_n - n}{\sqrt{n}} \Big|_{n \rightarrow \infty}$$

$$X_n \sim N(n/2, n/4)$$

$$\frac{2X_n - n}{\sqrt{n}} \sim N(0, 1) = Z$$

$$\mathbb{E} \left[\frac{2X_n - n}{\sqrt{n}} \right] = \frac{1}{\sqrt{n}} (2 \times n/2 - n) = 0$$

$$\mathbb{V} \left(\frac{2X_n - n}{\sqrt{n}} \right) = \frac{1}{n} \times 4 \times n/4 = 1$$

$$S_t = S_0 \exp(\mu t + \sigma \sqrt{t} \cdot Z)$$

$$S_t \sim \text{lognormal}(\mu t, \sigma^2 t)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

$\delta t^{>1}$ → ignored

under Q

$$q = \frac{e^{r\delta t} - d}{u - d} = \frac{e^{r\delta t} - e^{\mu\delta t - \sigma\sqrt{\delta t}}}{e^{\mu\delta t + \sigma\sqrt{\delta t}} - e^{\mu\delta t - \sigma\sqrt{\delta t}}}, \quad \delta t \rightarrow 0$$

$$= \frac{(1 + r\delta t) - (1 + \mu\delta t - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)}{(1 + \mu\delta t + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t) - (1 + \mu\delta t - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)}$$

$$= \frac{\delta t(r - \mu - \frac{1}{2}\sigma^2) + \sigma\sqrt{\delta t}}{2\sigma\sqrt{\delta t}}$$

$$= \frac{1}{2} \left(1 - \sqrt{\delta t} \left(\frac{\mu - r + \frac{1}{2}\sigma^2}{\sigma} \right) \right)$$

$$q = \frac{1}{2} \left(1 - \sqrt{\delta t} \left(\frac{\mu - r + \frac{1}{2} \sigma^2}{\sigma} \right) \right), \quad X_n \sim \text{Binom}(n, q)$$

$$S_t = S_0 \exp \left(\mu t + \sigma \sqrt{t} \cdot \left(\frac{2X_n - n}{\sqrt{n}} \right) \right), \quad \mathbb{E}(X_n) = nq$$

$$\mathbb{V}(X_n) = nq(1-q)$$

$$\mathbb{E} \left[\frac{2X_n - n}{\sqrt{n}} \right] = \frac{1}{\sqrt{n}} (2nq - n) = \frac{1}{\sqrt{n}} \left(-n \sqrt{\delta t} \left(\frac{\mu - r + \frac{1}{2} \sigma^2}{\sigma} \right) \right) = -\sqrt{t} \left(\frac{\mu - r + \frac{1}{2} \sigma^2}{\sigma} \right)$$

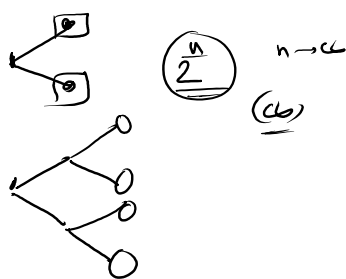
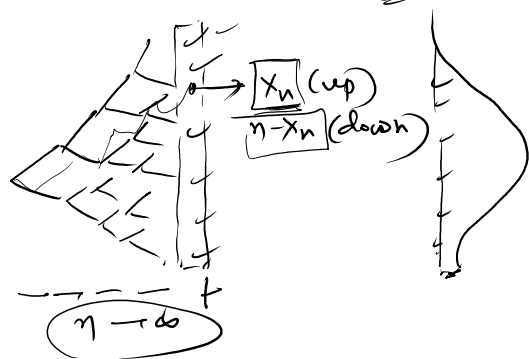
$\delta t \rightarrow 0$

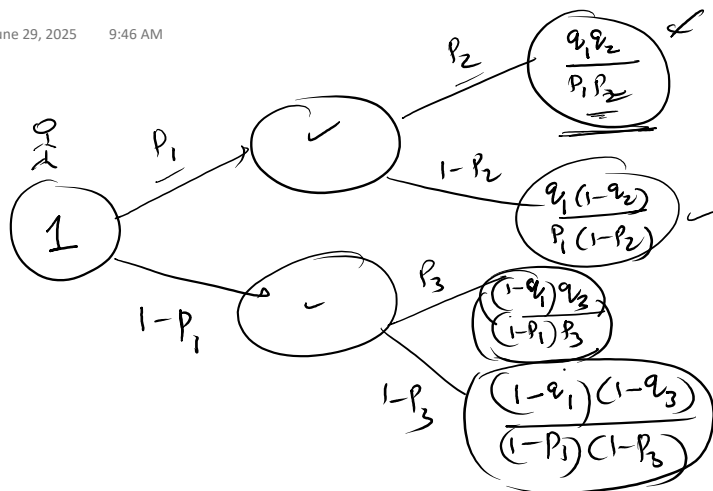
$$\mathbb{V} \left(\frac{2X_n - n}{\sqrt{n}} \right) = \frac{1}{n} (4nq(1-q)) = 1 \quad (\text{Homework}) \quad X = \mu + \sigma Z$$

$$S_t = S_0 \exp \left(\mu t + \sigma \sqrt{t} \left(-\frac{\sqrt{t}}{\sigma} \left(\mu - r + \frac{1}{2} \sigma^2 \right) + 1 \cdot Z \right) \right)$$

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma \sqrt{t} \cdot Z \right)$$

$ds/s = r dt + \sigma dw$





$$\underline{\underline{\zeta_t = \mathbb{E}^P(\underline{\underline{\zeta_t}} | \mathcal{F}_t)}}$$

$$\underline{\underline{\zeta_0 = \mathbb{E}^P(\underline{\underline{\zeta_t}} | \mathcal{F}_0)}}$$

$$q_1q_2 + q_1(1-q_2) + (1-q_1)q_3 + (1-q_1)(1-q_3) = q_1 + q_3 - q_1q_3 + 1 - q_1 - q_3 + q_1q_3 = 1$$

$$\underline{\underline{\zeta_1 = \mathbb{E}^P(\underline{\underline{\zeta_t}} | \mathcal{F}_1)}}$$

$$\zeta_2 = \mathbb{E}^P(\zeta_t | \mathcal{F}_T) = \zeta_2$$